Roll No. | | | | | | | |

SET-1

Even End Semester Examination 2025

Name of the Course: B. Tech Biotechnology — Semester: 4th

Name of the Paper: Bioprocess Engineering — Paper Code: **TBT-403**

Time: 3 Hours — Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Each sub question carries 10 marks

Q1	(20 marks)	
(a)	Compare and Contrast Graphical method and Richards rapid method applied for del estimation.	
(b)	Evaluate the effectiveness of the Plackett-Burman design in optimizing a complex medium for a biopharmaceutical fermentation. Justify your answer.	**CO1**
(c)	Define bioprocess engineering and briefly describe its historical development. List the different methods of medium sterilization.	
Q2	(20 marks)	
(a)	Derive an expression for total biomass in Fed Batch culture and doubling time in Batch culture kinetics.	
(b)	A 500 L chemostat is used for the continuous production of ethanol by a bacterium. The feed contains 120 g/L of glucose, and the flow rate is 50 L/h. The following kinetic parameters are given: $\mu max = 0.4\ h^{-1}$, Ks = 1.5 g/L, Yp/s = 0.45 g ethanol/g glucose and Yx/s = 0.2 g biomass/g glucose Calculate: 1.The dilution rate (D) 2.The steady-state ethanol concentration 3.The volumetric ethanol productivity (DP)	**CO2**
(c)	A fed-batch culture is used to produce an extracellular enzyme. The process starts with an initial volume of 100 L in the bioreactor, containing a biomass concentration of 5 g/L and negligible product concentration. A substrate solution with a concentration of 200 g/L is fed into the reactor at a constant flow rate of 8 L/h. The fermentation is carried out for 12 hours. The following parameters are given: Yx/s (biomass yield coefficient) = 0.4 g biomass/g substrate, Yp/x (product yield coefficient) = 10 U/g biomass, μmax = 0.2 h^-1 and Ks = 1 g/L Assume that substrate is primarily consumed for biomass and product formation, and there is no product initially in the reactor. Also, assume that the specific growth rate is much greater than the dilution rate.	

	1.The final volume of the culture (2) The final biomass concentration (3) The total amount of enzyme produced.	
Q3	(20 marks)	**CO3**
(a)	Derive an expression for calculation of KLa (Volumetric oxygen transfer coefficient) using Dynamic Gassing out technique?	
(b)	How do cell concentration, Temperature, Bubble size and anti foaming agents affect oxygen transfer from gas bubble to cell site in fermenter?	
(c)	A continuously stirred tank reactor with a working volume of 500 L is used for fermentative production. At steady state, the cell concentration is 10g/L and the specific oxygen uptake rate is 0.42 g O_2/ g cells.h. The inlet air has an oxygen partial pressure of 0.32 atm. The Henry's law constant for oxygen in the liquid medium is H= 0.05 atm. L/ g O_2. The Dissolved oxygen concentration in the Bioreactor is maintained at 40% of saturation. Estimate the value of KLa .	
Q4	(20 marks)	**CO4**
(a)	Analyze the advantages and disadvantages of solid-state fermentation compared to submerged fermentation.	
(b)	Choose an antibiotic and briefly outline the steps involved in its bioprocess production.	
(c)	Evaluate the importance of bioprocesses in the production of recombinant insulin and other biopharmaceuticals. Discuss the key challenges and future directions.	
Q5	(20 marks)	**CO5**
(a)	Explain the principle of biosensors. Describe the recent applications of 5 enzymes in the biopharmaceutical industry.	
(b)	How do you immobilize cells using active and passive immobilization methods?	
(c)	Analyze the working of Amperometric Biosensors. Highlight their applications.	